ChatGPT Conversation Report

Lab 4 — Vibe Coding
Math Flashcards Repair Lab

Project	Math Flashcards Repair Lab
Technology	Python and Pygame
Assigned repository	***https://github.com/narendrakathare03/PES2UG25CS814_SE_Lab_04***

1. Purpose

This document summarizes the main prompts, debugging steps, implementation decisions, and testing discussed while repairing the Math Flashcards game with ChatGPT. It is an editable summary, not a verbatim export of every message. If the instructor requires the complete literal chat history, include the original ChatGPT conversation export or share link as well.

2. Lab Objective

Fix the deliberate arithmetic calculation bug and implement three features: a per-question countdown timer, consecutive-correct streak multipliers, and integer division that always produces a whole-number answer.

3. Conversation and Implementation Log

Task 1 — Fix the calculation bug

Student request: The function returned int(f"{self.num_a}{self.num_b}"). Explain and fix the bug.

Assistant guidance: The expression joined the operands as text instead of calculating them. The calculation function was updated to evaluate +, -, and * according to the active operator.

Example tests: 4 + 5 = 9; 9 - 3 = 6; 6 * 7 = 42.

Result: The arithmetic calculation bug was fixed.

Task 2 — Add a countdown timer bar

Student request: Add a visible timer bar below the flashcard and register a missed attempt when time expires.

Assistant guidance: A 10-second timer was initialized. pygame.time.get_ticks() was used to calculate elapsed time, and a bar was drawn in render(). The timer resets for each new question, and timeout handling advances to another card.

Result: The timer bar and timeout behavior were implemented and observed in the running game.

Task 3 — Add streak multipliers

Student request: Increase rewards for consecutive correct answers and reset the streak after a wrong answer or timeout.

Assistant guidance: The code introduced streak and multiplier variables. The multiplier is capped at 4, awarding 1, 2, 3, then 4 points for consecutive correct answers. Wrong answers and timeouts reset the streak and multiplier. Empty submissions are handled separately.

Result: The game displayed streak information and increased score rewards.

Task 4 — Add integer division

Student request: Add division questions with whole-number answers and no remainder.

Assistant guidance: A divisor and quotient are generated first, then the dividend is set to divisor × quotient. This prevents zero divisors and guarantees exact division. The expected-answer function uses integer division (//).

Example: 28 / 7 = 4.

Result: Integer division was added to card generation and answer calculation.

Debugging — Feedback message mismatch

Student request: The current flashcard and green feedback message showed different calculations. Review the full code.

Assistant guidance: The feedback describes the question just answered, while the card has already advanced. The previous operands and operator were saved before generating the next card, and those saved values were used in the feedback message.

Result: The feedback was corrected to refer to the question that was submitted.

Final code review

Student request: Check the complete game_engine.py and provide a corrected version.

Assistant guidance: The review identified that input and timer resets were incorrectly nested under the subtraction condition, and division returned a float. The resets were moved to the end of card generation, integer division was used, and timeout handling was guarded to prevent repeated counting.

Result: A complete corrected version of game_engine.py was provided and the user confirmed that it was working.

4. Key Code Patterns

Arithmetic evaluation

```
if self.operator == "+":
    return self.num_a + self.num_b
elif self.operator == "-":
    return self.num_a - self.num_b
elif self.operator == "*":
    return self.num_a * self.num_b
elif self.operator == "/":
    return self.num_a // self.num_b
```

Integer-safe division generation

```
self.num_b = random.randint(2, 12)
quotient = random.randint(2, 12)
self.num_a = self.num_b * quotient
```

Streak multiplier

```
self.streak += 1
self.multiplier = min(self.streak, 4)
points = self.multiplier
self.score += points
```

5. Verification Checklist

Feature	What to verify
Arithmetic bug	Addition, subtraction, and multiplication accept correct answers.
Timer bar	The bar decreases and a timeout advances to the next card.
Streak multiplier	Consecutive correct answers increase rewards up to 4×; wrong answers/timeouts reset the streak.
Integer division	Division questions produce whole-number answers without remainders.
Feedback	The success message displays the operands from the question just answered.

6. Conclusion

The Math Flashcards Repair Lab was completed through an iterative AI-assisted debugging and development process. The arithmetic bug was fixed, and the countdown timer, streak multipliers, and integer division were implemented. The user confirmed that the corrected game was working. Before submission, include the original before video, the after video, updated source code, and the complete conversation export or share link as required.